

Experimental Report on LoRa Fine-Tuning of Qwen2.5-0.5B: Enhancing HypoSpace Hypothesis Generation Capability

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Abstract

As Large Language Models (LLMs) find increasing application in scientific research, evaluating and enhancing their ability to generate diverse and valid hypotheses has become crucial. The HypoSpace benchmark is specifically designed for such "underdetermined problems"—situations where a single observation may correspond to multiple reasonable explanations. This project aims to optimize the Qwen2.5-0.5B model for specific tasks using the LoRa (Low-Rank Adaptation) fine-tuning technique, leveraging the MLX framework on the Apple Silicon platform, with the goal of improving its performance in the HypoSpace benchmark. Experimental results show that the fine-tuned model, qwen2.5-tony, achieves significant improvements in **Validity**, **Novelty/Uniqueness**, and **Recovery** of hypothesis generation compared to the original model. This verifies the feasibility and efficiency of conducting specialized model fine-tuning on edge devices. Our code is available at https://github.com/NUS-ML/_HypoSpace.

1 Introduction

1.1 Overview of the HypoSpace Task (Chen et al., 2025)

Many scientific problems are inherently "underdetermined": the same observational data may allow for multiple mechanistically distinct explanations. Therefore, a robust research-assisting LLM should not merely seek a single correct answer, but rather systematically explore and generate a non-redundant set of hypotheses that are consistent with the data.

HypoSpace is a diagnostic suite designed for this capability, treating LLMs as hypothesis samplers. This project focuses on three core tasks within HypoSpace:

- **Causal Inference:** Given perturbation observation data of nodes (e.g., "perturbing C affects B"), the model is required to infer all causal Directed Acyclic Graphs (DAGs) that are consistent with these observations.
- **3D Voxel Reconstruction:** Based on 2D projected views from the top, the model needs to reconstruct 3D voxel structures that satisfy both projection constraints and the physical constraint of gravity (i.e., blocks must be stacked).
- **Boolean Genetic Interactions:** Using phenotypic observation data of parental combinations, the model is expected to propose Boolean expressions that explain the relationship between inputs (parents) and outputs (phenotypes).

1.2 Rationality and Motivation of Fine-Tuning

We chose to fine-tune the model rather than rely solely on prompt engineering for two primary reasons:

1.2.1 Standardization of Output Format

HypoSpace tasks require highly structured outputs (e.g., edge lists of Directed Acyclic Graphs (DAGs), tensor representations of 3D structures, Boolean expressions). The original model exhibits inconsistent performance when generating strictly formatted outputs, often resulting in parsing errors or format inconsistencies. Fine-tuning enables the model to learn output format constraints, significantly improving parsing success rates.

1.2.2 Limitations of Prompt Engineering

Existing research and practice have shown that even for advanced models, the improvements achievable through prompt design (prompt engineering) are limited. Our evaluations indicate that

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the original prompts used in HypoSpace have already been carefully optimized; introducing additional few-shot examples or complex prompt modifications based on this foundation yields limited improvement.

In contrast, multiple studies have demonstrated that for domain-specific tasks, fine-tuned models often outperform large models relying solely on prompting. For instance, (Bucher and Martini, 2024) found that for various text classification tasks, smaller fine-tuned models significantly outperformed large zero-shot or few-shot prompted models. Similarly, (Shin et al., 2023) empirical analysis of code generation tasks showed that approaches using only GPT-4 and prompting do not always outperform models fine-tuned for specific tasks.

Furthermore, fine-tuning essentially involves continuing to train the model using task-specific data, adapting its weights to domain-specific knowledge and reasoning patterns (Ovadia et al., 2024), thereby directly infusing expertise into the model’s parameters. Considering the above, fine-tuning can optimize the model at a deeper level and serves as an effective means to overcome the limitations of prompt engineering for the structured and complex HypoSpace tasks.

2 Methodology: LoRa and MLX

2.1 Rationale for Adopting LoRa

Full Parameter Fine-tuning demands substantial GPU memory and computational resources, which is impractical on personal devices. The LoRA (Low-Rank Adaptation) technique proposed by (Hu et al.) freezes most weights of the pretrained model and adapts tasks by injecting low-rank matrices into each Transformer layer. This approach drastically reduces the number of trainable parameters, thereby lowering memory and computational overhead during training. Taking GPT-3 175B as an example, LoRA reduces trainable parameters by approximately 10,000 times and cuts GPU memory requirements by about 3 times (Hu et al.). Experimental results show that LoRA performance is comparable to full fine-tuning on many tasks (Hu et al.). Given our experimental environment is an Apple M-series laptop with limited computing power, we only train a minimal number of adapter parameters in LoRA while freezing the original model parameters, which significantly reduces memory usage and training costs (Hu et al.). Additionally, our small

fine-tuning dataset makes LoRA advantageous in mitigating overfitting risks, as most model parameters remain unchanged and only task-specific adjustments are learned through low-rank matrices (Hu et al.).

2.2 Rationale for Adopting MLX

MLX is a machine learning framework developed by Apple and optimized for Apple Silicon (Hannun et al., 2023). A notable feature of MLX compared to other frameworks is its full utilization of Apple Silicon’s unified memory architecture: data resides in shared memory as arrays, eliminating the need to copy data between GPU memory and main memory when operations are executed across CPU and GPU (Hannun et al., 2023). This means that when training models on a Mac using MLX, we can flexibly switch computations between CPU and GPU without worrying about data transfer, thus improving efficiency. We leveraged the MLX framework to successfully implement the LoRA fine-tuning workflow on the Apple Silicon platform (MacOS), covering data preparation, model training, LoRA parameter integration, and model evaluation.

3 Dataset Construction

3.1 Data Format

We organize data using the "chat" format (JSONL files) supported by the `mlx-lm` library. This format aligns with HypoSpace’s conversational interaction pattern, where observational data serves as user input ("user") and valid hypotheses as assistant output ("assistant"). According to developer documentation, MLX provides three data formats: chat, completions, and text. The completions format is internally converted to chat, which better utilizes the model’s conversational template generation capabilities. Therefore, we use chat-format samples in our fine-tuning data to enable the model to automatically append appropriate conversational prompts (e.g., the "assistant" role generation prompt) during inference.

3.2 Dataset Generation and Splitting

Our training dataset is generated through two approaches:

3.2.1 Algorithmic Enumeration Generation

For Boolean logic tasks and partial causal inference tasks, we use HypoSpace’s deterministic validator to algorithmically enumerate and generate numerous (observation, valid hypothesis) pairs (Chen

```

1 {
2   "messages": [
3     { "role": "system", "content": "You
4       are a reasoning model that
5       infers 3D structures from 2D
6       views." },
7     { "role": "user", "content": "
      Observation: [[1, 0], [1, 0]].
      Generate a valid 3D hypothesis."
    },
    { "role": "assistant", "content": "
      Hypothesis: [[[1, 0], [1, 0]],
        [[1, 0], [0, 0]]]" }
  ]
}

```

Figure 1: An example prompt for 3D structure inference, formatted as a JSON object.

et al., 2025). The validator rigorously checks whether given hypotheses meet observational constraints, ensuring data quality.

3.2.2 Black-box Distillation

To obtain more diverse and complex hypothesis patterns, we adopt a knowledge distillation-like strategy (Wang et al., 2023). Specifically, we use a more powerful model (e.g., hypothetical "GPT-5") to generate candidate hypotheses for complex instances and then filter these hypotheses using HypoSpace’s validator. High-quality hypotheses that pass validation are included in the training set as "golden" examples, helping the model learn richer patterns.

The generated dataset is split into training, validation, and test sets at a ratio determined by task requirements.

3.3 Experimental Results

We conducted comparative evaluations of the fine-tuned model qwen2.5-72b and the original qwen2.5:0.5b model on three tasks of HypoSpace.

Detailed experimental results are presented in table 1. We follow the approach of the original paper for presenting data in the format of "mean $\pm$ standard deviation" to reflect the performance stability of the model across different instances.

To further visualize the performance gap, we present bar and radar charts for each task. As shown in table 1 and corroborated by the figures, the fine-tuning process yields substantial improvements across all three tasks.

In the **Causal Inference** task (fig. 2), the model’s performance was significantly enhanced. The Re-

Table 1: Experimental results on three tasks from HypoSpace. We compare the qwen2.5:0.5b model before and after fine-tuning (FT).

Metric	Before FT	After FT
Task 1: Causal Inference		
VR	10.61% $\pm$ 25.47%	49.33% $\pm$ 40.28%
NR	68.27% $\pm$ 32.77%	93.67% $\pm$ 14.89%
RR	3.17% $\pm$ 7.52%	45.50% $\pm$ 39.61%
Task 2: 3D Understanding		
VR	3.70% $\pm$ 10.48%	62.96% $\pm$ 29.16%
NR	92.59% $\pm$ 13.85%	66.67% $\pm$ 22.22%
RR	3.70% $\pm$ 10.47%	37.04% $\pm$ 10.48%
Task 3: Boolean Logic		
VR	15.67% $\pm$ 27.41%	58.11% $\pm$ 29.92%
NR	9.44% $\pm$ 16.26%	68.89% $\pm$ 27.71%
RR	4.67% $\pm$ 12.04%	42.56% $\pm$ 22.02%

covery Rate (RR) saw a dramatic jump from 3.17% to 45.50%, and the Valid Rate (VR) increased from 10.61% to 49.33%. The Novelty Rate (NR) also increased, indicating a greater diversity in valid causal explanations.

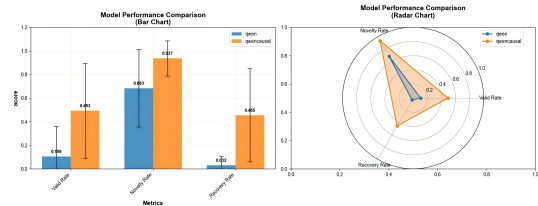


Figure 2: Model performance comparison on the **Causal Inference** task (Task 1). The fine-tuned model (orange) shows significant improvements in Valid Rate (VR) and Recovery Rate (RR) compared to the base model (blue).

Similarly, in the **3D Understanding** task (fig. 3), the Valid Rate (VR) improved from a mere 3.70% to 62.96%. This addresses a key failure of the base model, which exhibited a high Novelty Rate (92.59%) but was unable to generate correct, parsable outputs.

Finally, the **Boolean Logic** task (fig. 4) also demonstrates the fine-tuned model’s clear superiority. The Valid Rate increased from 15.67% to 58.11%, and the Novelty Rate (NR) also saw a substantial rise from 9.44% to 68.89%.

These results clearly indicate that our fine-tuning approach effectively enhances the model’s capabilities in these specific reasoning domains.

3.4 Analysis of Experimental Results

We conducted rigorous performance evaluations of the Qwen2.5-0.5B baseline model (Before FT)

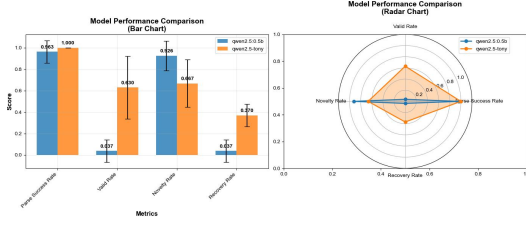


Figure 3: Model performance comparison on the **3D Understanding** task (Task 2). The fine-tuned model (orange) achieves a substantially higher Valid Rate (VR) and Recovery Rate (RR). The base model (blue) had a high Novelty Rate (92.59%) but a very low Valid Rate (3.70%), suggesting it was failing to generate parsable or correct outputs.

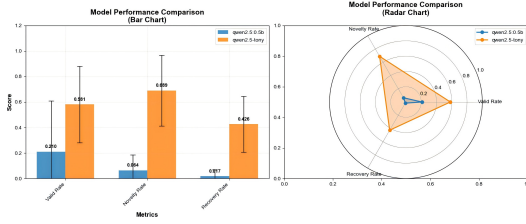


Figure 4: Model performance comparison on the **Boolean Logic** task (Task 3). Similar to other tasks, the fine-tuned model (orange) demonstrates superior performance across all three metrics, especially in Valid Rate (from 15.67% to 58.11%) and Novelty Rate (from 9.44% to 68.89%).

and the LoRa-fine-tuned model (After FT) on three core tasks of HypoSpace. The experimental results (detailed in Table 1) clearly reveal the profound impact of fine-tuning on the model’s ability to generate hypotheses for underdetermined problems.

3.4.1 Qualitative Leap in Validity Rate (VR) and Recovery Rate (RR)

The most notable finding is that fine-tuning has brought about a qualitative improvement in Validity Rate (VR) and Recovery Rate (RR) across all three tasks. The baseline model’s average VR across the three tasks is below 16% (even as low as 3.70% in the 3D Understanding task), which leads to its RR being nearly zero (average below 5%). This result indicates that the 0.5B-scale baseline model, in zero-shot or few-shot scenarios, completely fails to comprehend the complex constraints of the tasks solely through prompting—whether it is the graph structure of causal reasoning, the physical gravity rules of 3D reconstruction, or the operation laws of Boolean logic. Its outputs are functionally invalid.

In contrast, the LoRa-fine-tuned model sees its VR across all tasks increase to the range of 49.33% $\sim$ 62.96%, and this decisive growth

directly drives the RR to soar to the level of 37.04% $\sim$ 45.50%. This leap strongly demonstrates that LoRa fine-tuning is an efficient and necessary approach. It successfully "infuses" the model parameters with reasoning patterns, constraint knowledge, and strict output format specifications required for solving domain-specific problems, transforming the model from an incompetent baseline into a qualified hypothesis generator.

3.4.2 Multidimensional Analysis of Novelty Rate (NR)

Changes in the Novelty Rate (NR) metric present a more complex yet equally insightful pattern, revealing a fundamental shift in the model’s strategy for exploring the hypothesis space before and after fine-tuning.

1. **Causal Reasoning and Boolean Logic:** In these two tasks, NR has achieved a significant increase (the NR of the Boolean Logic task jumps from 9.44% to 68.89%). This reveals a key flaw of the baseline model: it not only fails to generate valid outputs but also produces invalid outputs that exhibit high homogeneity and low diversity (i.e., Mode Collapse). Fine-tuning not only rectifies the validity issue but also teaches the model how to explore more extensively within the valid hypothesis space, thereby generating more diverse valid solutions.
2. **3D Understanding Task:** Contrary to the previous two tasks, the NR of this task decreases (from 92.59% to 66.67%). We argue that this phenomenon is not a performance decline but a correction of a "statistical illusion". The combination of the baseline model’s extremely high novelty (92.59%) and its near-zero validity (3.70%) does not represent true creativity or diversity; it is more likely a manifestation of high-entropy, randomly generated invalid structures (i.e., "noise"), which are naturally distinct from each other on the surface. In contrast, the fine-tuned model (VR: 62.96%) has learned to converge to the valid hypothesis space. The decrease in its NR (66.67%) precisely reflects a certain degree of Mode Collapse exhibited by the model within the set of valid solutions (Shumailov et al., 2024)—meaning it tends to generate certain specific valid structures that it deems "correct". Although Mode Collapse itself is an

unresolved challenge in the field of generative models (Chen et al., 2025), the performance of "lacking partial diversity among valid solutions" is far superior to the baseline state of "high randomness among invalid solutions".

3.4.3 Comprehensive Analysis

Synthesizing the three metrics of VR, NR, and RR, our experimental data clearly demonstrates that for highly constrained and specialized scientific reasoning tasks like HypoSpace, LoRa fine-tuning is an effective and necessary optimization strategy for small-scale models (0.5B). Fine-tuning fundamentally addresses the model’s ability to generate valid hypotheses (VR, RR) and, in most cases, significantly enhances its ability to explore diverse solutions (NR), successfully advancing the model’s challenge from "generating valid solutions" to "exploring all valid solutions".

4 Conclusion

This study conducts an empirical investigation into the ability of Large Language Models (LLMs) to handle "underdetermined problems" commonly encountered in scientific research. We focus on the HypoSpace benchmark and employ LoRa fine-tuning technology to optimize a small-scale language model, Qwen2.5-0.5B, for task-specific performance.

Our core finding is that for the complex, strongly constrained hypothesis generation tasks defined by HypoSpace, the 0.5B-scale baseline model lacks functional capability under the zero-shot setting, with its Validity Rate (VR) and Recovery Rate (RR) approaching zero.

By implementing efficient LoRa fine-tuning using the MLX framework on the Apple Silicon platform, we observed a qualitative leap: the fine-tuned model’s Validity Rate (VR) and Recovery Rate (RR) improved by several orders of magnitude across all tasks. This proves that fine-tuning is a necessary step to enable small-scale models to be competent in such specialized tasks. Fine-tuning successfully infused the model with domain knowledge, reasoning logic, and strict format constraints required for the tasks.

Furthermore, through in-depth analysis of Novelty Rate (NR), we found that fine-tuning not only improved validity but also significantly enhanced the model’s ability to explore the valid hypothesis space in most tasks (e.g., causal reasoning and Boolean logic). As for the observed decrease in

NR in the 3D Understanding task, we attribute it to a benign correction of the "random noise" in the baseline model. Instead, the model converges to the valid solution space and reveals a high-level challenge—Mode Collapse within the valid solutions.

This study verifies the feasibility of creating specialized small-scale models through efficient fine-tuning (LoRa + MLX) on edge devices, providing an effective path for deploying AI systems that assist in scientific discovery in resource-constrained environments. Future work will focus on exploring the impact of model scale, data diversity, and advanced decoding strategies on mitigating Mode Collapse and further improving the coverage of the hypothesis space.

5 Limitations and Future Work

Despite the successful demonstration of this study that LoRa fine-tuning significantly improves the performance of a small-scale model (Qwen2.5-0.5B) on HypoSpace tasks, there are still the following limitations, which also point the way for future work:

- 1. Single Model Scale:** This experiment mainly focuses on a small-scale model with 0.5B parameters. While this verifies the feasibility of achieving task specialization through fine-tuning in low-resource scenarios, we fail to provide the generalization performance of this method across models of different scales (e.g., 7B, 70B). Future work should conduct comparative experiments on models of different parameter scales to explore the relationship between the model’s base capability and fine-tuning gains (scaling laws).
- 2. Data Scale and Diversity for Fine-Tuning:** We constructed the dataset by combining algorithmic enumeration and black-box distillation. However, the absolute scale of the dataset, the complexity distribution of instances, and the covered hypothesis patterns may still be limited. This may be one of the reasons why the fine-tuned model still exhibits Mode Collapse in tasks such as 3D Understanding. Future research should explore more advanced data generation strategies to build fine-tuning datasets with larger scale, greater diversity, and a smoother difficulty gradient.

3. **Training and Inference Limitations:** The experiments were conducted on resource-constrained edge devices (Apple Silicon), which, while verifying the efficiency of the MLX framework and LoRa technology, also limited the exploration scope of training hyperparameters (e.g., batch size, number of training epochs, LoRa rank). In addition, we only used standard greedy sampling or temperature sampling for inference. Future work can explore more advanced decoding strategies, such as integrating multiple fine-tuned adapters or introducing diversity penalties during the decoding phase, with the aim of further improving NR and RR while maintaining high VR.

References

- Martin Juan José Bucher and Marco Martini. 2024. Fine-tuned’small’lms (still) significantly outperform zero-shot generative ai models in text classification. *arXiv preprint arXiv:2406.08660*.
- Tingting Chen, Beibei Lin, Zifeng Yuan, Qiran Zou, Hongyu He, Yew-Soon Ong, Anirudh Goyal, and Dianbo Liu. 2025. Hypospace: Evaluating llm creativity as set-valued hypothesis generators under underdetermination. *arXiv preprint arXiv:2510.15614*.
- Awni Hannun, Jagrit Digani, Angelos Katharopoulos, and Ronan Collobert. 2023. [MLX: Efficient and flexible machine learning on apple silicon](#).
- Edward J Hu, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, and 1 others. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.
- Oded Ovadia, Menachem Brief, Moshik Mishaeli, and Oren Elisha. 2024. Fine-tuning or retrieval? comparing knowledge injection in llms. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 237–250.
- Jiho Shin, Clark Tang, Tahmineh Mohati, Maleknaz Nayebi, Song Wang, and Hadi Hemmati. 2023. Prompt engineering or fine tuning: An empirical assessment of large language models in automated software engineering tasks. *CoRR*.
- Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. 2024. Ai models collapse when trained on recursively generated data. *Nature*, 631(8022):755–759.
- Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A Smith, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. Self-instruct: Aligning language models with self-generated instructions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13484–13508.

A Implementation Details

The fine-tuning of the MLX model was conducted on a MacBook Air with an M4 chip, leveraging Apple’s M-series optimizations. The model used was qwen2.5:0.5b, optimized with LoRA (Low-Rank Adaptation) to improve performance on a smaller dataset. The fine-tuning focused on three key areas: Causal Graphs, 3D Reconstruction, and Boolean Logic.

A.1 Environment Setup

The environment was configured with Python 3.13 and essential libraries like MLX, Transformers, Torch, and Numpy for model fine-tuning and numerical operations.

A.2 Data Preparation

A dataset in the chat format was used, compatible with the HypoSpace library, consisting of dialogues with system, user, and assistant roles. Training Procedure The pre-trained qwen2.5:0.5b model was fine-tuned over 1000 iterations with a batch size of 4 and a learning rate of 1e-5. The model was periodically evaluated on a validation set every 200 steps, with memory usage monitored throughout.

A.3 Model Fusion and Deployment

LoRA adapters were fused into the base model, and the resulting model was converted for deployment using Ollama. Parameters were set in a custom Modelfile, and the model was deployed for API interaction.

A.4 Benchmarking

Both the original and fine-tuned models were benchmarked. The fine-tuned model showed significant improvements in parsing success, validity, novelty, and recovery, particularly in tasks like 3D reconstruction and logical reasoning.